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ARTICLE

Loneliness is associated with more trust but worse trustworthiness expectations

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Abstract
Subjective feelings of loneliness emerge due to unsatisfactory social relationships, representing a major risk for mental and physical well-being. Despite its social nature, evidence on how loneliness affects social behaviours and expectations is lacking. Using Bayesian analyses and economic games, we show in three different studies that lonelier individuals trusted their partners to a greater extent despite less favourable trustworthiness expectations, showing a greater discrepancy between their trusting behaviours and their expectations of others' trustworthiness. Such discrepancy was reversed in extravert individuals who also reported to be less lonely. These results provide evidence on two opposing effects of loneliness as a motivator for social connections and promoter of social withdrawal, and demonstrate the moderating role of personality traits. This work contributes to a better understanding of how loneliness impacts social behaviour and social expectations, with important downstream clinical implications for varying health conditions associated with heightened feelings of loneliness.

KEYWORDS
Bayesian analysis, extraversion, investment game, loneliness, trust, trust game, trustworthiness expectations

BACKGROUND

> Yet the sweet converse of an innocent mind,
>
> Whose words are images of thoughts refin'd,

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Is my soul's pleasure

O Solitude! John Keats

Loneliness is formally defined as feelings of perceived social isolation triggered by unsatisfying social relationships that make people feel being isolated and left-out (Cacioppo et al., 2000). As such, loneliness partially diverges from feelings of objective social isolation or social exclusion, which are more tightly linked to an individual's objective social embeddedness (de Jong Gierveld & Havens, 2010).

Feelings of loneliness have been associated with depression, anxiety, mental and cardiovascular diseases, representing a major risk for all-cause mortality (Ernst & Cacioppo, 1999; Hawkley et al., 2003; Hawkley & Capitanio, 2015). Together with living alone, loneliness increases the chances of dying by around 30% even after controlling for age, sex and overall health conditions (Holt-Lunstad et al., 2015). Importantly, loneliness induces social withdrawal, which has been recognized as a transdiagnostic symptom common to the most frequent psychiatric disorders in the European Union (Porcelli et al., 2019).

Even though previous studies have investigated different aspects of loneliness (e.g., physical, psychological, and neural), little is known about its dynamics from a social perspective. On the one hand, loneliness has been proposed to function as an internal warning signal that alerts individuals about their unsatisfactory social relationships and promotes behaviours that facilitate connection and bonding (Cacioppo, Grippo, et al., 2015) by activating a 'reaffiliation motive' for the development of meaningful and supporting relationships (Perlman & Peplau, 1981; Qualter et al., 2015). With this respect, loneliness is a central mechanism to the promotion of companionship and intimacy, and to the evolution of adaptive responses aimed at remedying the discrepancy between the preferred and realized levels of social connection (Cacioppo, Cacioppo, et al., 2015).

On the other hand, lonely individuals have been characterized by an implicit hypervigilance for social threats (Cacioppo et al., 2014; Pickett & Gardner, 2005), as they pay more attention to threating interactions, are more aware of negative evaluations from others and remember more negative feedbacks (Duck et al., 1994; Gardner et al., 2005; Pickett & Gardner, 2005). Given their less supportive social network, such hypervigilance is supposed to protect them from life-threatening events, but has the undesired consequence of promoting social withdrawal (Cacioppo & Hawkley, 2009).

However, empirical evidence on these mechanisms underlying loneliness is still lacking. In particular, as beliefs about others are central to how individuals behave in social interactions, it is of pivotal importance to elucidate the effects of loneliness on socially relevant expectations of others and how these effects reverberate in the concomitant social behaviours.

LONELINESS AND TRUST

Despite trust being central to different social behaviours, no studies have so far investigated the relationships between trusting behaviours and loneliness. An early study using the Prisoner's Dilemma found that lonely adolescents did not increase their cooperation in response to their partner's cooperative behaviour (Rotenberg, 1994), suggesting that lonely adolescents might need more social cues to trust a partner for cooperation. Further, lower trust beliefs in same-gender peers in 10-year-old children have been associated with both greater feelings of loneliness and less cooperative behaviour (Rotenberg et al., 2004). However, explicit measures of trusting behaviour were lacking in these studies. Similarly, little is known on the relationships between loneliness and trustworthiness expectations. Previous studies have investigated the impact of loneliness on subjective beliefs in others' expectations of one's own trustworthiness, suggesting that lonely individuals more strongly believe that their peers do not trust them, but these results are not conclusive (Rotenberg, 1994; Rotenberg et al., 2004, 2010). As trustworthiness expectations of one's partner are central to trusting behaviours (Chaudhuri & Gangadharan, 2007; Cox & Deck, 2007), loneliness-dependent modulations of individual expectations of others' trustworthiness might play a central role in an individual's willingness to trust.

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Importantly, trusting behaviours often diverge from one's expectations of others' trustworthiness. For instance, individuals seem to trust more than what their trustworthiness expectations would predict, likely due to the fact that expectations of a partner's trustworthiness are only one factor influencing trust decisions (Aimone & Houser, 2013; Ashraf et al., 2006; Fetchenhauer & Dunning, 2009). In addition, trusting behaviour is shaped by reputational concerns related to attempts to signal one's own trustworthiness and willingness to cooperate (Jordan, Hoffman, Bloom, & Rand, 2016; Jordan, Hoffman, Nowak, & Rand, 2016), by the desire to comply with a fairness norm (Falk et al., 2008), and by respect for the partner (Dunning et al., 2014). Other studies have further suggested that trust decisions in the classic investment game are not only determined by social factors related to the trust dimension but also by simpler attempts to reduce inequalities when resource distributions between players are unequal (Cox, 2002, 2004). Moreover, a divergence between trustworthiness expectations and trusting behaviour is supported by evidence that trusting individuals are not themselves trustworthy towards their trusting partners (Chaudhuri & Gangadharan, 2007).

LONELINESS AND PERSONALITY TRAITS

Because loneliness emerges as a consequence of social contacts of low quality, personality traits that facilitate contact and connections with others might moderate the impact of loneliness on social expectations and social behaviours (Soto & John, 2017). In particular, extraversion plays a role in positive affect (Lucas & Fujita, 2000), sociability (John et al., 2008) and happiness (Tan et al., 2018; Tkach & Lyubomirsky, 2006)—psychological dimensions that promote a more positive outlook on social participation and social partners (Watson & Clark, 1997). Extravert individuals are less sensitive to social threats (Costa & McCrae, 1992; Wilt & Revelle, 2017), which likely contributes to alleviate the hypothesized hypervigilance in lonely individuals and to help them form more positive social expectations. Both early (Hojat, 1982; Saklofske & Yackulic, 1989) and more recent studies (Lampraki et al., 2019; Mund & Neyer, 2018) have found lower levels of loneliness in more extravert individuals. Reduced levels of loneliness have been observed in both clinical (e.g., depression) and healthy extravert adults (Itzick et al., 2020; Schutter et al., 2020; von Soest et al., 2020), as well as extravert adolescents (Cheng & Furnham, 2002).

Similarly, agreeableness has been associated with better intimate relationships (Graziano et al., 1996; Jensen-Campbell et al., 2002) and more helpful (David & Suls, 1999; Watson & Clark, 1992) and cooperative behaviours (Schroeder et al., 2015). Consequently, agreeable individuals exhibit prosocial tendencies and behaviours (Graziano & Eisenberg, 1997; Perunovic, 2007), which likely help them connect and protect them from feelings of loneliness. In a sample of adolescents from 11 to 21 years old, high levels of loneliness were associated with low agreeableness, low self-esteem, less depressive moods and reduced compulsive internet use (van der Aa et al., 2009). Further, adolescents with chronically high levels of loneliness report the lowest levels of agreeableness over a 5-year period (Vanhalst et al., 2013). Higher agreeableness has also been found associated with lower levels of loneliness in short-term divorced adults (Lampraki et al., 2019) and in a longitudinal study, lower levels of agreeableness predicted greater loneliness in old men (but not women) 5 years later (Ormstad et al., 2020). Higher levels of agreeableness are, finally, associated with less loneliness not only in healthy individuals but also in schizophrenia patients (Lindner et al., 2014).

Hence, given the role that such individual personality traits (i.e., agreeableness and extraversion) play in people's propensity to sociability and social connections (Buecker et al., 2020; Soto & John, 2017), the relationship between loneliness and social behaviours/expectations might further differ on the individual level as a function of those personality traits. In particular, the relationship between loneliness and trustworthiness expectations should be less pronounced for individuals with higher levels of extraversion and agreeableness, while, as more extravert and agreeable individuals are already more socially engaged, the effects of loneliness on trust should be stronger for individuals with lower levels of extraversion and agreeableness.

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CURRENT RESEARCH

Here, in three consecutive studies using Bayesian predictive models and economic games, we investigated the effects of feelings of loneliness on trusting behaviours and trustworthiness expectations. Two different predictions can be made. On the one hand, if loneliness fosters prosocial behaviours that help establish social relationships and bonds, lonely individuals should manifest high levels of trust. On the other hand, if loneliness induces a negativity bias towards others, they should have more negative expectations of others' trustworthiness. In particular, we employ the one-shot investment game (IG), which simulates a social situation in which trusting behaviours and trustworthiness expectations of social partners diverge (Dunning et al., 2014; Fetchenhauer & Dunning, 2009). This allows us to investigate the opposite effects of loneliness on social expectations and behaviours by testing whether loneliness widens the mismatch between trusting behaviours and trustworthiness expectations. Finally, the impact of loneliness on one's own trust and expectations of others' trustworthiness might further be moderated by traits (i.e., extraversion and agreeableness) that have been shown to be associated with a stronger social engagement or a more positive outlook on life.

STUDY 1

In Study 1, we tested whether loneliness is associated with less favourable expectations of others' trustworthiness. To further test whether loneliness is also associated with a greater willingness to trust, we measured participants' trusting behaviour using the one-shot IG (Berg et al., 1995). In the IG, a player (the investor) receives an initial endowment and decides whether to share some of it with a partner (the trustee). The amount shared by the investor is multiplied (usually tripled) and passed on to the trustee who decides whether to share any portion of the multiplied amount back to the investor. The IG has been used to study trust in reciprocity (Camerer, 2003; Chaudhuri et al., 2002; Krueger et al., 2007, 2008), as the investor's decision is regarded as a measure of trust in the trustee's reciprocity (Chaudhuri & Gangadharan, 2007; Csukás et al., 2008).

In particular, we used the one-shot version of the IG where investors and trustees engage in a single interaction. Here, the investor does not know the trustee and receives no information about the trustee's personality, reputation or likely behaviour. This economic game is well suited as investors' decisions in the one-shot IG are not only a function of their general expectations of the partner's trustworthiness but also represent a manifestation of their own general prosocial tendency (Cesarini et al., 2008; Fehr & Gintis, 2007; Hahn et al., 2015; Kanagaretnam et al., 2009), attempts to signal their own trustworthiness and willingness to cooperate (Jordan, Hoffman, Bloom, & Rand, 2016; Jordan, Hoffman, Nowak, & Rand, 2016), and compliance with social norms of fairness and respect for the partner (Dunning et al., 2014; Falk et al., 2008). Hence, the one-shot IG naturally creates a social situation where trusting behaviour and trustworthiness expectations of a social partner partly diverge, thereby allowing us to test whether the degree to which the two diverge can be explained by an individual's levels of loneliness.

Methods

Subjects

In Study 1, a sample of 209 participants (163 females; *age* ± *SD*, 22.6 ± 3.83) took part in the experiment. The sample size was based on the effect sizes of the estimates for trust and trustworthiness expectations from the Bayesian model of the main study (see below) fitted to pilot data predicting loneliness scores. Standardized parameter estimates were 0.18 ($SD = .13$) for trusting behaviour and −0.21 ($SD = .14$) for trustworthiness expectations. As to reach a power of 80% with these effect sizes around 180 participants were necessary, we set out to recruit approximately 200 participants. The study was ran online

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using SoSci Survey (Leiner, 2019), was made available to users via www.soscisurvey.de, and advertised at the University's internal mailing list. The study was approved by the local Ethics Committee of the University. All participants provided written informed consent for participation. The experiment lasted around 30 min, and participants received up to €10 for participation. Three participants had missing data and were hence excluded from analyses. All studies were conducted in accordance with the Declaration of Helsinki. All study materials have been made publicly available under the following link. Participants sequentially played the investment game and dictator game in random order, and filled out questionnaires and biographical questions at the end of the experiment (see Materials).

Materials

Self-report measures

Subjective feelings of loneliness were measured using the UCLA loneliness scale (Russell et al., 1980; Windisch & Kniel, 1988), which has 20 items such as: 'I feel shut out and excluded by others' or 'People are around me but not with me'. Participants rate each item on a 5-point Likert-scale ranging from 'I often feel this way' to 'I never feel this way'. Previous results indicate that the measure is highly reliable, both in terms of internal consistency (coefficient α ranging from .89 to .94) and test–retest reliability over a 1-year period ($r = .73$) (Russell et al., 1980). In our sample from Study 1, Cronbach's α for the UCLA loneliness scale was .92, indicating excellent internal consistency. Expectations of others' trustworthiness were measured using the preference survey module for trust preferences (Falk et al., 2016), which consisted of 6 items like 'Most people will respond in kind when they are trusted by others' and 'Most people are trustworthy'. Ratings were on a 5-point Likert-scale from 'strongly disagree' to 'strongly agree'. Cronbach's α was .86, indicating good internal consistency.

In addition, we acquired data on control variables that have been show to play a role in the severity of loneliness, like subjective feelings of happiness, self-esteem, momentary feelings of loneliness and need to belong, as well as biographical variables (Bellucci, 2020; Cavallo et al., 2014; Steptoe & Fancourt, 2019; von Soest et al., 2020). Happiness was assessed with the 4 items of the subjective happiness scale (Lyubomirsky & Lepper, 1999). Previous research has shown good reliability for the subjective happiness scale with Cronbach's *a* values ranging from .79 to .94 (Lyubomirsky & Lepper, 1999). In our sample from Study 1, Cronbach's *a* for the happiness scale was .89, indicating good internal consistency. Self-esteem was assessed using the 10 items (e.g., 'I take a positive attitude towards myself') of the self-esteem scale (Rosenberg, 1965; von Collani & Herzberg, 2003). Previous research found good reliability for the scale with a Cronbach's alpha of .89 (von Collani & Herzberg, 2003). In Study 1, Cronbach's *a* was .91, indicating excellent internal consistency. Momentary feelings of loneliness were determined by using a 3-item loneliness scale (Hughes et al., 2004; Waytz et al., 2015). Previous research has reported acceptable reliability with a Cronbach's alpha of .72 (Hughes et al., 2004). Consistent with previous results, in Study 1, Cronbach's *a* was .72. Finally, participants' need to belong relied on the 10 items (e.g., 'I want other people to accept me') of the need to belong scale (Leary et al., 2013). In Study 1, Cronbach's *a* was .74, indicating acceptable internal consistency.

Biographical variables were introduced in all models and controlled for age, education, sex (1 = female, 0 = male), economic status (from low to high social class on a scale from 0 to 100), relationship status (1 = single, 0 = in relationship/married), and living situation (1 = living alone, 0 = living in a shared flat, with family or partner).

Behavioural tasks

For trusting behaviour, the one-shot IG was employed (Berg et al., 1995). The IG is a two-person sequential economic game in which one person has the role of investor and the other of trustee. In our study, the investor receives an initial endowment of 10 monetary units and decides whether to share any portion of it with the trustee. If the investor decides to share, the shared amount is tripled and passed on to the trustee who decides whether to share any portion of the received amount back to the investor. As we were interested

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in studying trusting behaviour and the investor's decision is thought to capture the investor's trust in the trustee, while the trustee's decision is considered as the trustee's reciprocity to the investor's trust, our participants always played as investor. However, at the beginning of the experiments they were told that roles in the game were randomly assigned. Also, participants knew that the amount they earned in the task would have been converted into real money and paid out to them at the end of the experiment as bonus payment. So, they knew that their decisions were not hypothetical but had real impact.

Since as previously mentioned, the investor's decision in the one-shot IG might also be motivated by other reasons than expectations of trustworthiness, a second economic game (i.e., the dictator game) was used to control for variance in participants' decisions that is not strictly related to the trust dimension but are, for example, associated with other factors such as generosity (Forsythe et al., 1994; Kahneman et al., 1986). Like the IG, in the dictator game, a player, the dictator, receives an initial endowment (in our study, 10 monetary units) and needs to decide whether to share any portion of it with another player, the receiver. Unlike the IG, amounts shared by dictators are not multiplied and receivers do not have any decision to make. Previous empirical investigations have shown that even though dictators do not have any advantage from sharing and there are no deterrents for not sharing, most individuals share some portion of their initial endowment, suggesting that it captures general generosity and compliance with social norms (Forsythe et al., 1994; Grossmann et al., 2020). Previous work has provided evidence that combining the investment and dictator games helps identify an individual's willingness to trust (Cox, 2002, 2004), as transfers in the dictator game can be used to control for other-regarding preferences (e.g., generosity) that affect behaviour in the IG. One distinct feature of transfers in the IG as resulting from such compound motives is that they are consistently higher than transfers in the dictator game. This was the case also in our sample from Study 1 where transfers in the IG ($M = 5.95$, $SD = 2.71$) were significantly higher than transfers in the dictator game ($M = 4.69$, $SD = 2.48$; $t_{208} = 6.52$, $p < .00001$; Table S1) with the Bayes Factor (BF) pointing to very strong evidence for the alternative hypothesis ($BF_{01} > 1.5 \times 10^7$). Hence, following this tradition, our participants played both games in the role of investor and dictator and an individual's degree of trust was represented by the difference between the amount shared in the games. However, to control for game and role effects (Johnson & Mislin, 2011), our participants were not informed at the beginning of the experiment that they were going to play both games but learnt about each of them only sequentially, and the order in which they played the games was randomized across participants.

Statistical analyses

We used Bayesian regression models to predict individual loneliness levels and personality traits in all three studies. We considered linear regression models with Student's t errors, because compared to normal distributions, t distributions have heavier tails and treat observations far from the regression line as high-variance observations, yielding results similar to those obtained by downweighting outliers (Gelman et al., 2014). A random variable Y is said to follow a Student's t distribution if its density function is given by:

$$StudentT(y \mid \nu, \mu, \sigma) = \frac{\Gamma\left(\frac{\nu+1}{2}\right)}{\sigma\sqrt{\nu\pi}\,\Gamma\left(\frac{\nu}{2}\right)}\left(1 + \frac{(y-\mu)^2}{\nu\sigma^2}\right)^{-\frac{\nu+1}{2}}. \tag{1}$$

A linear regression with Student's t errors can be written as:

$$y_t = \beta_1 + x_{1,t}\beta_2 + \ \dots\ + x_{k,t}\beta_k + \varepsilon_t \tag{2}$$

where $(\varepsilon_t \mid \lambda_t, \sigma^2) \sim \mathcal{N}(0, \sigma^2\lambda_t)$, and λ_t is a latent variable that scales the variance of Y and has an inverse-gamma distribution, such as $\lambda_t \sim IG\left(\frac{\nu}{2}, \frac{\nu}{2}\right)$. Further, $x_{1,t} + \ \dots\ + x_{k,t}$ are the k regressors and

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$\beta_1 + \ldots + \beta_k$ are the associated regression coefficients over observations $t = 1, \ldots, T$. In Study 1, the regression model predicting individual loneliness scores had 12 regressors. Six regressors including the two variables of interest, that is, trusting behaviour in the IG and trustworthiness expectations, and four other measurements of psychological traits such as happiness scores, self-esteem, momentary feelings of loneliness, and need to belong. The other six regressors entailed biographical variables like education, economic status, sex, age, relationship status and living condition.

We fitted the models in Stan (https://mc-stan.org) using the brms (Bürkner, 2017) package in R, which employs the no-U-turn sampler for efficient exploration of posterior estimates. In particular, we ran 4 chains with 21,000 iterations each and 1000 burn-in samples. Posterior point estimates represented the expected value of the posterior distributions of models' parameters. Uncertainty in the estimation of the models' parameters was represented by 95% highest posterior density intervals (HPDI). To build the model fitted to the data from Study 1 and Study 2, we used uninformative normally distributed priors for the predictors ($\mathcal{N}(\mu, \sigma^2)$ where $\mu = 0$ and $\sigma^2 = 10$). After estimation, posterior predictive checks using in-sample predictions based on the parameters' posterior distributions were performed to evaluate how well the model represented the fitted data. To further test the model's generalizability, posterior point estimates of models' parameters evaluated from Study 1 and 2 were used to generate out-of-sample predictions on data from Study 3. Posterior estimates from the models fitted to the data from Study 1 and 2 were used as priors for the models fitted to the data from Study 3. Finally, bivariate associations were tested using Spearman correlations (ρ values).

Results and discussion

First of all, we investigated the opposite effects of loneliness on social expectations and behaviours by testing whether the degree to which trustworthiness expectations and trusting behaviour diverged can be explained by individual feelings of loneliness. We observed that lonelier individuals trusted more despite having more negative trustworthiness expectations than less lonely individuals ($\rho_{204} = .37$, $p < .0001$; Figure 1d), indicating an increased discrepancy between one's own trust and trustworthiness expectations for higher levels of loneliness. To better characterize the effects of loneliness on trust and trustworthiness expectations, and to control for other factors that might explain these relationships, we ran Bayesian regression analyses with trust and trustworthiness expectations as separate regressors predicting individual loneliness scores together with other control variables (see Table 1). We confirmed that loneliness entertained opposing and significant relationships with both trusting behaviour and trustworthiness expectations (see Figure 1a). Greater feelings of loneliness were associated with higher trust ($\beta = 0.51$, 95% credible intervals (95% CI) = [0.16; 0.87]) but less favourable expectations of others' trustworthiness ($\beta = -0.16$, 95% CI = [−0.26; −0.05]), suggesting that lonelier individuals trusted more despite expecting others to be less trustworthy.

Importantly, using posterior point estimates of the fitted Bayesian model, we found that our model well-predicted feelings of loneliness in both in- and out-of-sample participants (Figure 1). Pointwise posterior predictions of individual loneliness levels highly correlated with the observed loneliness scores in Study 1 ($r_{204} = .77$, $p < .0001$; Figure 1b), and the model fitted to Study 1 well-predicted loneliness scores of out-of-sample individuals from Study 3 ($r_{215} = .55$, $p < .0001$; Figure 1c).

These results indicate that lonelier individuals are indeed prone to trust their partners to a greater extent than less lonely individuals but also that they have less favourable expectations of their social partners. We also checked whether these results would hold even when looking only at transfers in the IG separated from transfers in the dictator game. In particular, given that our measure of trust is the result of the difference between the sharing behaviours in the IG and in the dictator game, it is not clear whether the positive relationship between loneliness and our measure of trust was really due to higher transfers during the IG in lonely individuals. To test this, we ran an additional Bayesian regression model with sharing behaviour in the IG as predictor of individual feelings of loneliness and transfers

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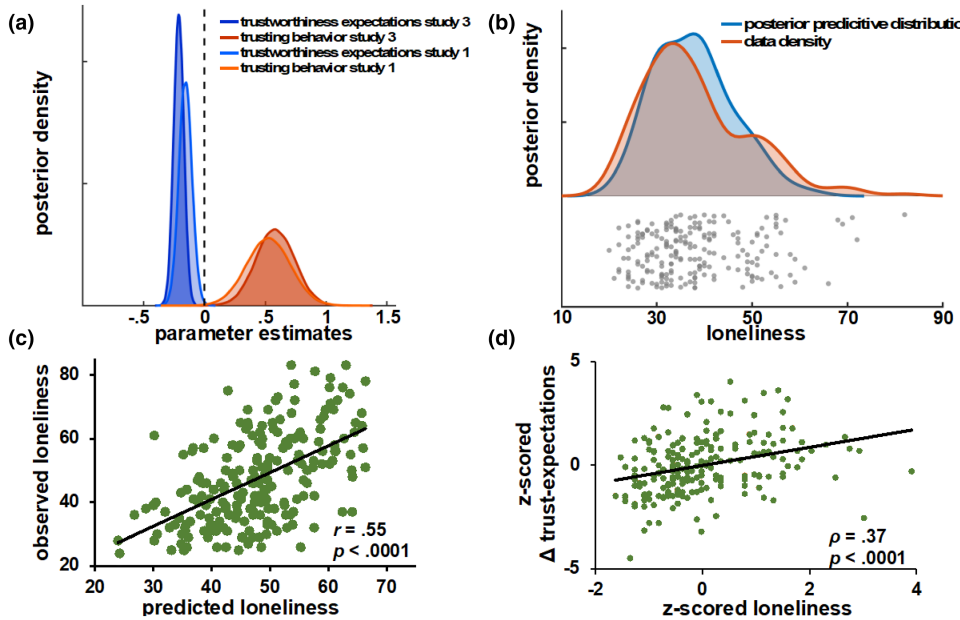

FIGURE 1 Loneliness is negatively associated with trustworthiness expectations but positively with trusting behaviour. (a) Loneliness was negatively associated with trustworthiness expectations but positively with trust in the one-shot investment game in both Study 1 and Study 3. (b) The posterior predictive distribution shows that the Bayesian model accurately captured the underlying distribution of our data. Depicted data points are individual loneliness scores. (c) The Bayesian model based on posterior point estimates fitted to loneliness data from Study 1 predicted with high accuracy loneliness levels of non-seen participants from Study 3. (d) Loneliness was associated with a bigger divergence between an individual's trust and trustworthiness expectations in Study 1.

in the dictator game as control variable. Results show that transfers in the IG were still positively associated with loneliness levels ($\beta = 0.44$, 95% CI = [0.02; 0.85]), even though this effect size was relatively smaller than the one found with our measure of trust, probably due to the fact that sheer transfers in the IG do not directly reflect individual's trust levels but rather a compound construct of trust and other preferences as previously argued (Cox, 2002, 2004).

Similar results were found when running two separate models with transfers in the IG and dictator game. The relationship between transfers in the IG and loneliness without controlling for transfers in the dictator game was still positive and similar to the findings observed above ($\beta = 0.45$, 95% CI = [0.08; 0.82]). Interestingly, the relationship between transfers in the dictator game and loneliness without controlling for IG transfers was negative ($\beta = -0.19$, 95% CI = [−0.58; 0.21]), suggesting that lonelier individuals start with lower baseline, other-regarding preferences and might overly compensate with strongly higher trust levels leading to higher IG transfers.

Further, loneliness was negatively associated with individual characteristics that might play a role in exacerbating such feelings of isolation (Table 1). In particular, lonelier individuals reported to be less happy ($\beta = -1.89$, 95% CI = [−3.08; −0.68]), have lower self-esteem ($\beta = -0.28$, 95% CI = [−0.47; −0.09]), and greater transient feelings of loneliness ($\beta = 3.41$, 95% CI = [2.60; 4.23]). These results are in line with previous work indicating that lonely individuals experience stronger dissatisfaction with oneself and one's own life, likely discouraging them from seeking out social contacts that might reduce their loneliness (Bellucci, 2020; Luhmann & Hennecke, 2017). As no associations were observed between loneliness and an individual's need to belong (Table 1), our data do not seem to support the hypothesis that lonely individuals feel lonely because they have a stronger need to be socially embedded or a stronger desire for acceptance. Also, in line with previous work, we observed that sex was strongly

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TABLE 1 Relationships of loneliness with trusting behaviour and trustworthiness expectations.

	Studies			
	Study 1		Study 3	
	Posterior point estimates	95% HPDI	Posterior point estimates	95% HPDI
Regressors				
Intercept	33.41	[21.67; 45.01]	51.22	[43.12; 59.26]
Trusting behaviour	0.51	[0.16; 0.87]	0.57	[0.26; 0.88]
Trustworthiness expectations	−0.16	[−0.26; −0.05]	−0.21	[−0.29; −0.13]
Happiness	−1.89	[−3.08; −0.68]	−2.67	[−3.54; −1.79]
Self-esteem	−0.28	[−0.47; −0.09]	−0.36	[−0.50; −0.22]
Momentary loneliness	3.41	[2.60; 4.23]	2.46	[1.91; 3.03]
Need to belong	0.08	[−0.16; 0.33]	0.22	[0.03; 0.41]
Biographical variables				
Education	1.63	[−0.54; 3.80]	0.58	[−0.41; 1.57]
Economic status	3.42	[−2.70; 9.64]	2.35	[1.48; 3.22]
Sex	−3.37	[−5.84; −0.95]	−2.81	[−4.67; −0.95]
Age	0.22	[−0.05; 0.49]	0.17	[0.03; 0.32]
Relationship status	1.35	[−0.68; 3.43]	2.25	[0.63; 3.89]
Living condition	1.74	[−0.51; 3.99]	0.76	[−0.95; 2.47]

Note: Posterior point estimates (mean) of posterior distributions from Bayesian models predicting individual loneliness levels in Study 1 and Study 3, as well as their 95% highest posterior density intervals (HDPI). Coding for sex: 1 = female, 0 = male; for relationship status: 1 = single, 0 = in relationship/married; and for living condition: 1 = living alone, 0 = living in a shared flat, with family or partner.

associated with loneliness (Table 1), as females reported lower levels of loneliness than males ($\beta = -3.37$, 95% CI = [−5.84; −0.95]).

Importantly, the original associations of loneliness with trustworthiness expectations and trusting behaviours were much stronger before the introduction of our control variables. In particular, posterior point estimates for trust and trustworthiness in a predictive model that entailed no further regressors were $\beta = 0.54$ (95% CI = [0.07; 1.02]) and $\beta = -0.30$ (95% CI = [−0.43; −0.17]), respectively. Intriguingly, in line with our previous work (Bellucci, 2020), the strength of the relationship between loneliness and trustworthiness expectations was the one mostly impacted by the introduction of our control variables in the predictive model. In particular, the strength of the relationship between loneliness and trustworthiness expectations decreased the most when the regressor for happiness was entered into the model, suggesting that the variance in feelings of loneliness explained by one's expectations of others' trustworthiness strongly covaries with the variance explained by happiness levels.

STUDY 2

Next, we turned to the individual factors that might have a moderating role in the relationship between loneliness and trustworthiness expectations. Previous studies have shown that extravert and agreeable individuals tend to be more deeply embedded in a meaningful and supportive social network. Not only are they more outgoing and have more friends, they leave better impressions with others and have themselves more positive evaluations of their partners (Cheng & Furnham, 2002; Graziano & Eisenberg, 1997; McAdams & Olson, 2010). The greater sociability of extravert and agreeable individuals

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might indeed play an important role in their ability to form social bonds and intimate relationships. Hence, extraversion and agreeableness might moderate the impact of loneliness on trustworthiness expectations with higher levels of these personality traits contributing to more positive social expectations in lonely individuals.

Methods

Subjects

In Study 2, we had a total of 159 participants (81 females; 23.2 ± 3.02). Pilot data suggested that to reach a power of 80% for the relationship between loneliness and extraversion a sample of 100 participants was necessary. The study was approved by the local Ethics Committee of the University and all participants provided written informed consent for participation. The experiments lasted on average 2 h and participants received €8/hr for participation.

Materials

Personality traits were measured using the Big Five Inventory (Goldberg, 1990, 1992). Trustworthiness expectations, feelings of loneliness, happiness and self-esteem were measured as in Study 1. In Study 2, Cronbach's α values for each questionnaire were as follows: UCLA loneliness scale: .91; trust preferences: .87; self-esteem scale: .90; subjective happiness scale: .88; Big Five Inventory: .69 (median value across subscales).

Statistical analyses

The same Bayesian regression analysis from Study 1 was implemented in Study 2 to investigate the relationships of loneliness with trustworthiness expectations and personality traits. Relationships between subscales of personality traits and the strength of model-based predictions of individual loneliness scores were tested using Pearson correlations.

Further, the role of personality traits (extraversion and agreeableness) in moderating the relationship between loneliness and trustworthiness expectations was tested with moderation analyses. Generally, a moderator is a third variable that affects the zero-order correlation between two other variables such that the relationship between these two other variables is either substantially reduced or reversed (Baron & Kenny, 1986). More formally, a moderator is represented by a significant interaction between a focal predictor and the moderating variable in predicting an outcome variable related to both the predictor and the moderator. In particular, a moderation model can be described with the following regression:

$$y_i = \beta_0 + \beta_1 x_i + \beta_2 m_i + \beta_3 x_i m_i + \varepsilon_i, \tag{3}$$

where participant i's responses y_i are linearly predicted by the regressor x_i and this predictive relationship is moderated by the moderator m_i. The parameter estimates β_1 and β_2 are related to the effects of the predictor and the modulator on the dependent variable y_i, respectively, while β_3 expresses the moderating effect of these two variables on the outcome variable. In Study 2, y_i was taken to be participants' trustworthiness expectations that were regressed onto their loneliness feelings (x_i). Personality traits were fit as moderators (m_i) of the relationship between loneliness and trustworthiness expectations. Bayesian regression models with Student's t errors were implemented to estimate the moderation effects (β_3) by running 4 chains with 21,000 iterations each and 1000

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burn-in samples. Posterior point estimates represented the expected value of the posterior distributions of models' parameters. Uncertainty in the estimation of the models' parameters was represented by 95% HDPI. Uninformative, normally distributed priors were used for the predictors ($\mathcal{N}(\mu, \sigma^2)$ where $\mu = 0$ and $\sigma^2 = 10$).

Results and discussion

Bayesian analyses revealed that both extraversion and agreeableness were associated with loneliness and trustworthiness expectations (Figure 3a,b and Table 2). Extravert ($\beta = -2.73$, 95% CI = [−3.67; −1.80]) and agreeable ($\beta = -1.32$, 95% CI = [−2.26; −0.38]) individuals reported to feel less lonely and to have more positive trustworthiness expectations of others (extraversion: $\beta = 2.66$, 95% CI [1.36; 3.96]; agreeableness: $\beta = 7.12$, 95% CI = [5.86; 8.38]). Posterior predictive checks indicate high levels of concordance of our model with the observed data for both loneliness ($r_{157} = .59$, $p < .0001$) and trustworthiness expectations ($r_{157} = .71$, $p < .0001$). Further, pointwise posterior predictions indicate that our model well-predicted feelings of loneliness ($r_{217} = .55$, $p < .0001$) and trustworthiness expectations ($r_{217} = .63$, $p < .0001$; Figure 4a) in out-of-sample individuals from Study 3.

Moreover, neuroticism (Table 2) was also found to be positively associated with loneliness ($\beta = 1.29$, 95% CI = [0.53; 2.08]), whereas no associations were found with trustworthiness expectations ($\beta = -0.73$, 95% CI = [−1.66; 0.21]). On the contrary, conscientiousness and openness were not associated with either variable of interest (Table 2). Furthermore, while lonelier individuals reported to be less happy ($\varrho_{157} = -.50$, $p < .0001$; Figure 2a) and have lower self-esteem ($\varrho_{157} = -.50$, $p < .0001$; Figure 2b), extraverts reported higher levels of both happiness ($r_{157} = .39$, $p < .0001$) and self-esteem ($r_{157} = .31$, $p < .0001$). Same results were found for agreeableness (happiness: $r_{157} = .34$, $p < .0001$; self-esteem: $r_{157} = .17$, $p = .032$).

Next, we investigated whether extraversion and agreeableness, which were the only two personality traits that were strongly associated with both loneliness and trustworthiness expectations, moderated the effects of loneliness on social expectations. A moderation analysis revealed that both extraversion and agreeableness significantly moderated the negative relationship between loneliness and trustworthiness expectations (Figure 5a). The moderating effect was negative for both extraversion ($\beta = -0.14$, 95% CI = [−0.24; −0.04]) and agreeableness ($\beta = -0.09$, 95% CI = [−0.16; −0.01]; Figure 5a). That is, the negative relationship between loneliness and trustworthiness expectations was stronger for individuals with lower levels of extraversion and agreeableness, suggesting that

TABLE 2 Relationships between loneliness and personality traits in Study 2.

	Dependent variables			
	Trustworthiness expectations		Loneliness	
Regressors	Posterior point estimates	95% HPDI	Posterior point estimates	95% HPDI
Intercept	5.14	[−2.76; 13.06]	41.29	[34.97; 47.68]
Extraversion	2.66	[1.36; 3.96]	−2.73	[−3.67; −1.80]
Agreeableness	7.12	[5.86; 8.38]	−1.32	[−2.26; −0.38]
Neuroticism	−0.74	[−1.89; 0.40]	1.29	[0.53; 2.08]
Conscientiousness	1.21	[−0.14; 2.58]	−0.73	[−1.66; 0.21]
Openness	−0.63	[−1.80; 0.54]	0.06	[−0.77; 0.89]

Note: Posterior point estimates (mean) of posterior distributions from Bayesian models predicting individual loneliness levels and trustworthiness expectations from the five personality traits of the Big Five Inventory in Study 2, as well as their 95% highest posterior density intervals (HDPI).

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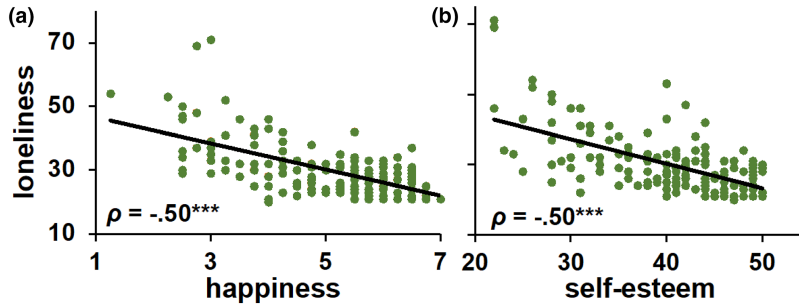

FIGURE 2 Relationships of loneliness with happiness and self-esteem. Loneliness was associated with lower happiness (a) and self-esteem levels (b) in Study 2. ***$p < .0001$.

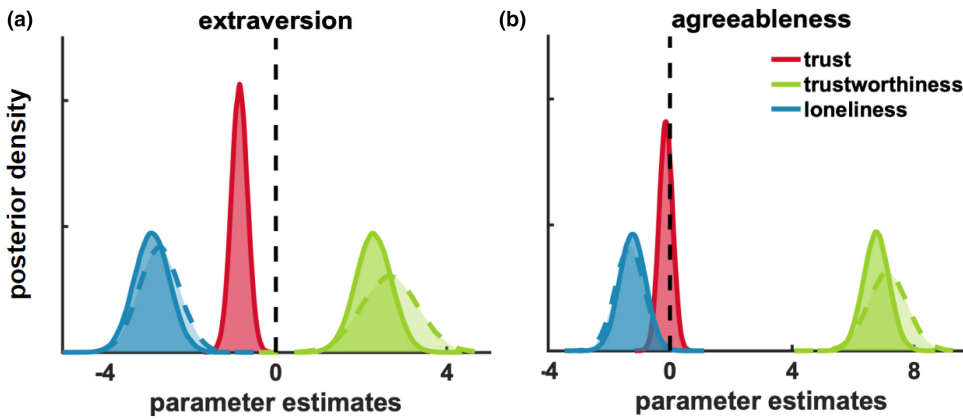

FIGURE 3 Relationships of personality traits with loneliness, trust and trustworthiness expectations. Posterior distributions over the parameters of the Bayesian models predicting trusting behaviour (red) individual loneliness levels (blue) and trustworthiness expectations (green) for extraversion (a) and agreeableness (b) from the five personality traits of the Big Five Inventory. In particular, extraversion can be seen to entertain a positive relationship with trustworthiness expectations but negative relationships with loneliness and trusting behaviour in both Study 2 (dotted lines) and Study 3 (solid lines). Vertical black dotted line separates the positive from the negative domain.

these personality traits were associated with a reduced effect of loneliness on expectations of others' trustworthiness. For completeness, we checked parameter estimates for extraversion ($\beta = 9.48$, 95% CI = [7.32; 11.58]) and agreeableness ($\beta = 10.74$, 95% CI = [8.87; 12.57]) in the moderation model, which were in line with the direction of the effects observed in the main regression analysis (Figure 3a,b and Table 2).

STUDY 3

In the previous studies, we observed that loneliness predicts a greater discrepancy between trust and trustworthiness expectations, as lonelier individuals trusted their partners more despite having more negative expectations of them. Further, personality traits such as extraversion and agreeableness moderated the negative relationship between feelings of loneliness and trustworthiness expectations. Next, we investigated whether these personality traits also moderate the positive relationship between loneliness and trusting behaviours observed in Study 1.

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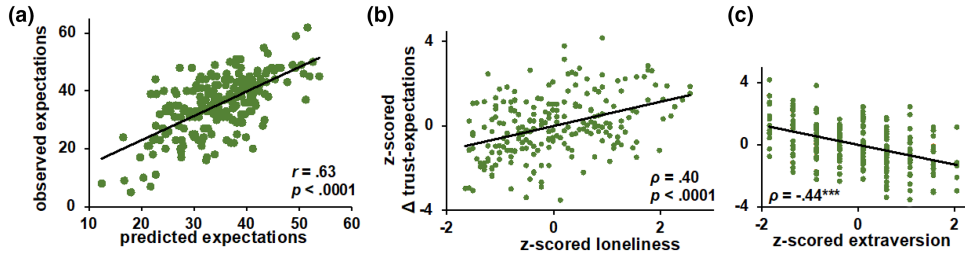

FIGURE 4 Divergence between trust and trustworthiness expectations was positively associated with loneliness and negatively with extraversion. (a) The Bayesian model based on posterior point estimates fitted to trustworthiness expectations from Study 2 predicted with high accuracy trustworthiness expectations by non-seen participants from Study 3. (b) Loneliness was associated with a bigger discrepancy between an individual's trust and trustworthiness expectations in Study 3. (c) On the contrary, extraversion was negatively associated with the discrepancy between an individual's trust and trustworthiness expectations in Study 3.

Methods

Subjects

In Study 3, a total of 219 participants (86 females; 28.3 ± 8.57) took part in the experiment. The sample size was based on the results of the relationships of loneliness with trusting behaviour and trustworthiness expectations from Study 1, indicating that 200 participants were necessary for a power of 80%. The study was run online using SoSci Survey and made available on Prolific (www.prolific.co), an online subject pool platform with a more diverse sample than University subject pools, allowing us to test our hypotheses on a more heterogeneous sample for better generalization of our results. The study was approved by the local Ethics Committee of the University. All participants provided written consent for participation. The experiment took around 30 min, and participants received €10 for participation. Two participants had missing data and were hence excluded from the analysis. Study procedures were as in Study 1.

Materials

Trusting behaviour was measured experimentally by using a combination of the one-shot investment and dictator games as in Study 1. As in Study 1, we checked whether transfers in the IG were higher than transfers in the dictator game in Study 3. We found strong evidence that this was the case also in our sample from Study 3 with transfers in the IG ($M = 5.38$, $SD = 2.85$) being significantly higher than transfers in the dictator game ($M = 4.63$, $SD = 2.84$; $t_{218} = 4.00$, $p < .0001$; $BF_{01} > 1.5 \times 10^2$; Table S1). Similarly, trustworthiness expectations and feelings of loneliness were also measured as in Study 1. Personality traits were measured with the Big Five Inventory as in Study 2. Finally, control variables that have been shown to be associated with feelings of loneliness (Bellucci, 2020; Cavallo et al., 2014; Steptoe & Fancourt, 2019; von Soest et al., 2020) were measured as in Study 1 and 2. In Study 3, Cronbach's α values for each questionnaire were as follows: UCLA loneliness scale: .93; trust preferences: .85; self-esteem scale: .90; subjective happiness scale: .85; momentary feelings of loneliness: .77; need to belong scale: .70; Big Five Inventory: .66 (median value across subscales).

Statistical analyses

The same Bayesian regression analyses as in Study 1 and 2 was implemented to investigate the relationships of loneliness with trusting behaviour, trustworthiness expectations and personality traits.

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TABLE 3 Relationships of personality traits with loneliness, trust and trustworthiness expectations in Study 3.

	Dependent variables					
	Trustworthiness expectations		Trusting behaviour		Loneliness	
Regressors	Posterior point estimates	95% HPDI	Posterior point estimates	95% HPDI	Posterior point estimates	95% HPDI
Intercept	7.28	[2.51; 12.09]	6.28	[3.56; 9.01]	51.95	[47.57; 56.34]
Extraversion	2.29	[1.46; 3.12]	−0.85	[−1.22; −0.48]	−2.90	[−3.72; −2.08]
Agreeableness	6.76	[5.92; 7.60]	−0.14	[−0.57; 0.29]	−1.23	[−2.07; −0.40]
Neuroticism	−0.43	[−1.15; 0.30]	−0.64	[−1.00; −0.29]	2.32	[1.63; 2.99]
Conscientiousness	1.08	[0.22; 1.94]	−0.08	[−0.48; 0.32]	−0.26	[−1.09; 0.57]
Openness	−0.56	[−1.33; 0.20]	−0.14	[−0.57; 0.29]	0.13	[−0.55; 0.81]

Note: Posterior point estimates (mean) of posterior distributions from Bayesian models predicting individual trustworthiness expectations, trusting behaviour and loneliness levels from the five personality traits of the Big Five Inventory in Study 3, as well as their 95% highest posterior density intervals (HDPI).

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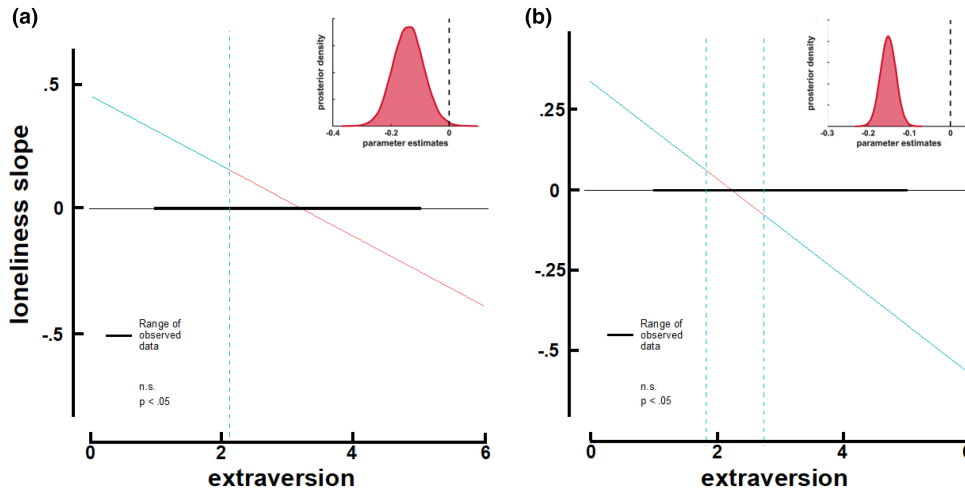

FIGURE 5 Moderation analyses with extraversion. Significant moderation effects of extraversion in Study 2 (a) and Study 3 (b) on the relationship between loneliness and trustworthiness expectations. Johnson–Neyman plots denote regions of significance in light blue and region of nonsignificance in red. Thick horizontal black lines show the range of observed data in the current studies. Vertical light blue dotted lines separate the regions of significance. In the insert, posterior parameter distributions of the respective interaction effects and vertical black dotted lines separate the positive from the negative domain.

Relationships between subscales of personality traits and the strength of model-based predictions of individual loneliness scores were tested using Pearson correlations.

Further, moderation analyses as in Study 2 were implemented to test the moderating role of personality traits in the relationships of loneliness with trusting behaviour and trustworthiness expectations. In Study 3, we ran different moderation analyses with participants' trustworthiness expectations and trusting behaviours in the one-shot IG as outcome variables (y_i).

Results and discussion

Replicating Study 1's results, the degree to which trusting behaviour and trustworthiness expectations diverged correlated with greater feelings of loneliness ($\rho_{217} = .40$, $p < .0001$; Figure 4b). We again observed that greater feelings of loneliness were associated with higher trust levels ($\beta = 0.57$, 95% CI = [0.26; 0.88]) but more negative trustworthiness expectations ($\beta = -0.21$, 95% CI = [−0.29; −0.13]; Figure 1a and Table 1). Like in Study 1, after controlling for transfers in the dictator game, the positive relationship between loneliness and our measure of trust was reflected by an increase in transfers in the IG in individuals with higher levels of loneliness ($\beta = 0.44$, 95% CI = [0.13; 0.75]). Again, replicating Study 1, we found opposite relationships between the separate transfers in the games and loneliness (IG: $\beta = 0.32$, 95% CI = [0.01; 0.63]; dictator game: $\beta = -0.32$, 95% CI = [−0.63; 0.01]), corroborating the hypothesis of a compensatory mechanism leading to higher trust levels in lonelier individuals.

We also replicated Study 2's results (Figure 3a,b and Table 3), as loneliness was again negatively related to extraversion ($\beta = -2.90$, 95% CI = [−3.72; −2.80]) and agreeableness ($\beta = -1.23$, 95% CI = [−2.07; −0.40]), while trustworthiness expectations were positively associated with extraversion ($\beta = 2.29$, 95% CI = [1.46; 3.12]) and agreeableness ($\beta = 6.76$, 95% CI = [5.92; 7.60]). Finally, in line with Study 1 (Table 1), lonelier individuals reported to be less happy ($\beta = -2.67$, 95% CI = [−3.54; −1.79]), and had greater transient feelings of loneliness ($\beta = 2.46$, 95% CI = [1.91; 3.03]). On the contrary, extravert and agreeable individuals reported both higher levels of happiness (extraversion: $r_{217} = .43$, $p < .0001$; agreeableness: $r_{217} = .35$, $p < .0001$) and self-esteem (extraversion: $r_{217} = .35$, $p < .0001$; agreeableness: $r_{217} = .25$, $p < .001$).

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Next, we turned to test the relationships between personality traits and trusting behaviour. We observed that more extravert individuals trusted their partners less ($\beta = -0.85$, 95% CI = [−1.22; −0.48]) despite having more positive expectations of others' trustworthiness ($\beta = 2.29$, 95% CI = [1.46; 3.12]; Table 3). This suggests that extraverts reported higher trustworthiness expectations than their levels of trust. Indeed, contrary to what we previously observed for loneliness, we found that extraversion was negatively correlated with the directional discrepancy between trusting behaviours and expectations of others' trustworthiness, pointing to more extraverted individuals reporting higher trustworthiness expectations than what their trusting behaviour would have suggested ($\varrho_{217} = -.44$, $p < .0001$; Figure 4c).

We then examined the moderating role of these personality traits. Replicating Study 2's results, extraversion and agreeableness significantly moderated the negative relationship between loneliness and trustworthiness expectations (Figure 5b). We found again evidence for a negative moderating effect of both extraversion ($\beta = -0.15$, 95% CI = [−0.19; −0.12]) and agreeableness ($\beta = -0.11$, 95% CI = [−0.14; −0.08]), with a stronger effect for extraversion (Figure 5b). These results indicate that higher levels of loneliness were more strongly associated with less favourable expectations of others' trustworthiness in individuals with lower levels of extraversion and agreeableness. In other words, lonelier individuals had more favourable expectations of others' trustworthiness the more extravert and agreeable they were. Again, for completeness we also report the main effects of extraversion ($\beta = 10.67$, 95% CI = [9.37; 11.97]) and agreeableness ($\beta = 11.42$, 95% CI = [10.29; 12.56]) in the moderation analysis, which were in the same direction as the ones observed in the main regressions of Studies 2 and 3 (Figure 3a,b and Tables 2 and 3) and similar in effect sizes as the moderation results of Study 2.

On the contrary, extraversion showed only a non-significant trend in moderating the relationship between loneliness and trusting behaviour ($\beta = 0.01$, 95% CI = [−0.009; 0.03]), which was in the opposite direction to the moderating effect of extraversion on the relationship between loneliness and trustworthiness expectations. This provides weak evidence that lonelier individuals with higher levels of extraversion manifest even higher trust levels during social interactions, that is, more aligned with their trustworthiness expectations. Moderating results for agreeableness were instead non-significant ($\beta = 0.001$, 95% CI = [−0.03; 0.03]). Again, the main effects of extraversion ($\beta = -1.19$, 95% CI = [−2.23; −0.14]) and agreeableness ($\beta = -.22$, 95% CI = [−1.63; 1.17]) in the moderation analysis were in the same direction as in Study 3 main regression model (Figure 3a,b and Table 3).

Finally, like in Study 1, we observed that females reported lower levels of loneliness than males ($\beta = -2.81$, 95% CI = [−4.67; −0.95]) and that singles ($\beta = 2.25$, 95% CI = [0.63; 3.89]), older adults ($\beta = 0.17$, 95% CI = [0.03; 0.32]) and people from higher economic status ($\beta = 2.35$, 95% CI = [1.48; 3.22]) reported higher loneliness levels (Table 1).

Also in Study 3, the original associations of loneliness with trustworthiness expectations and trusting behaviours were much stronger before the introduction of our control variables. Posterior point estimates for trust and trustworthiness in a predictive model that entailed no further regressors were $\beta = 0.59$ (95% CI = [0.17; 0.91]) and $\beta = -0.37$ (95% CI = [−0.50; −0.25]), respectively. Again, the strength of the relationship between loneliness and trustworthiness expectations was the one mostly impacted by the introduction of our control variables in the predictive model. In particular, the strength of the relationship between loneliness and trustworthiness expectations decreased the most when the regressors for happiness and self-esteem were entered into the model.

GENERAL DISCUSSION

Using economic games and Bayesian predictive models, we showed that higher levels of loneliness predicted a greater mismatch between social expectations and behaviours, with lonelier individuals trusting more despite more negative expectations of their partner's trustworthiness. These effects were moderated by personality traits associated with a more positive outlook on social interactions like extraversion. Notably, the effects of loneliness on trusting behaviours and expectations of others' trustworthiness were particularly strong for individuals with lower levels of extraversion.

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So far, evidence on the relationships of loneliness with social expectations and behaviours have remained underexplored. An early study observed that lonely adolescents do not increase their cooperation in response to a partner's cooperative behaviour, suggesting that lonely adolescents might need more cues about a partner's cooperativeness to engage in prosocial behaviours (Rotenberg, 1994). In more recent studies, lonelier individuals showed diminished responsiveness to a positive social interaction but indicated more positive prosocial attitudes (Bellucci, 2020; Lieberz et al., 2021). In the current work, lonelier individuals were indeed more prone to trust than less lonely individuals, concurring with the hypothesis that loneliness might act as a warning signal promoting behaviours that favour social connectedness and the formation of meaningful and supportive social bonds. In particular, given the role of trust in social relationships (Bellucci et al., 2020; Chang et al., 2010; De Neys et al., 2017), higher levels of trust in lonelier individuals might originate from attempts to increase social acceptance and inclusion (Krueger & Meyer-Lindenberg, 2019).

Concomitantly, our study shows that lonelier individuals are less likely to expect others to be trustworthy, which chimes with earlier evidence of a negativity bias in social evaluations in lonely individuals (Duck & Gilmour, 1981; Jones et al., 1981). This negativity bias has been suggested to reflect heightened vigilance for social threats (Cacioppo & Hawkley, 2009) and neuroimaging work has recently demonstrated enhanced recruitment of brain regions associated with an attentional brain network, likely underpinning such attentional bias (D'Agostino et al., 2018; Inagaki et al., 2015; Layden et al., 2017). Importantly, negative expectations in lonely individuals might play a role in self-isolating behaviours (Duck et al., 1994; Jones et al., 1983), as expectations of threating and harmful social interactions might motivate social withdrawal.

These findings highlight an adversarial interplay between the beneficial effects that loneliness has on an individual's tendency to connect and its deleterious impact on social expectations, pointing out how despite their negative expectations, our lonelier participants made themselves more vulnerable to their partners in trusting interactions. This discrepancy might represent the by-product of different mechanisms underlying both the reinforcement of negative expectations in lonely individuals and their willingness to engage in social interactions. For example, in line with previous work (Duck et al., 1994; Gardner et al., 2005), lonely individuals might attempt to connect with others to reduce their feelings of being isolated but might more negatively evaluate the social feedback they receive from others due to their more negative expectations. Such negative evaluations might ultimately confirm their negatively biased expectations and foster social withdrawal that exacerbates loneliness (Cacioppo & Hawkley, 2009). Such negativity bias might further be amplified by varying cognitive biases. A candidate is a memory bias that a recent study has shown to make lonely individuals remember more exclusion in the past and expect more exclusion in the future (Teneva & Lemay, 2020).

Another mechanism might be related to early evidence that lonely individuals are judged more negatively by others, their social skills are regarded as of poorer quality and elicit more social rejections (Borys & Perlman, 1985; Jones et al., 1982; Lieberz et al., 2021; Tsai & Reis, 2009). In line with this, the greater mismatch between lonely individuals' social behaviour and expectations might evoke negative impressions (e.g., of inauthenticity) in their interacting partners who might in turn respond with less trustworthy behaviours. These behaviours could then reinforce lonely individuals' initial, negative expectations, sparking off a spiral of uncooperativeness. Interestingly, the relationship of loneliness with this mismatch between an individual's trust and their trustworthiness expectations was reversed in more extravert individuals, suggesting that extroversion might actually help engage in less inauthentic behaviours. Future studies should explore the underlying mechanisms of these results. For example, since at zero acquaintance people expect others to behave less trustworthily than they actually do (Dunning et al., 2014), the lower degree of trust in extravert individuals might more closely match their partners' expectations, leading to a lower prediction error and belief uncertainty that might help build trust (Watson & Clark, 1997).

Finally, we observed lower levels of loneliness in females than males, while previous studies have reported higher loneliness levels in females (Bellucci, 2020; Cacioppo et al., 2009; Vanhalst et al., 2012). Gender differences in the social acceptability of self-disclosure about loneliness might provide insights

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into these mixed results. Indeed, our findings concord with recent evidence that in an implicit measure of loneliness, males report higher levels of loneliness than females (von Soest et al., 2020). These findings point to the possibility that the observed gender difference might partly be explained by internalized sex roles expectations, such as that it is less acceptable for men to admit they are lonely than for women (Lau & Kong, 1999) and that they might be more harshly excluded and stigmatized if they do so (Borys & Perlman, 1985). Even though we did not use any implicit measure of loneliness in our studies, the anonymity of the online experiments might have allowed for a more authentic disclosure in our male sample.

One might argue that a possible limitation of the current work is the absence of a manipulation of feelings of loneliness, to which we would like to highlight two points. First, feelings of loneliness are notoriously difficult to manipulate, as they refer to a subjective psychological state that is hardly accessible with objective measures. As prolonged social isolation might eventually elicit feelings of loneliness, a recent study has attempted to isolate participants for a prolonged period of time (i.e., 10 h), which, however, just led to a very small increase in social craving (Tomova et al., 2020). Second, we believe that loneliness should be treated as a preclinical symptom predictive of the onset of clinical disorders like depression and anxiety (Mann et al., 2017). Hence, loneliness should be investigated like any other clinical disorder by using a combination of questionnaire-based assessments and ecological, social paradigms, eventually with the help of stratified sampling (Miller & Chapman, 2001), instead of attempting to experimentally manipulate it.

Another potential limitation is related to the artificial context of the online studies, which did not imply a direct, interpersonal relationship between the participants. However, we tried to counteract that by telling participants that they were going to play with other participants online, which is a practice that should be fairly familiar to participants especially to those from Prolific. Moreover, previous studies have shown that participants' behaviour in the lab does not significantly differ from online behaviours in economic paradigms (Arechar et al., 2017). Lastly, the anonymity of the online context might also have had an impact on participants' behaviour and self-reports. However, at least with respect to self-reports on loneliness, this anonymity might actually have led to a more authentic disclosure in our male sample—as noted above.

In conclusion, this work does not only reveal the multifaceted relationships of loneliness with personality and social behaviours and expectations, but it also makes an additional interdisciplinary contribution by intersecting personality psychology and behavioural economics (Borghans et al., 2008; Engelmann et al., 2019). Building on the present findings, future research could for instance employ economic decision tasks like the ones used in the current study in combination with additional methodologies (e.g., vignette and surveys), experimental procedures (e.g., sequential paradigms) and computational models to test the degree to which lonely individuals are inclined to pay costs in attempts to build up meaningful relationships, how they learn from others' responses to their behaviour and how they remember others' past decisions. As one of the strongest predictor of depression, loneliness might be associated with computational learning patterns similar to those revealed in depressed individuals (Frey et al., 2019) with higher weights on negative information from social feedback that acts as a reinforcer of their negative expectations of others, thereby hindering their ability to form stable, meaningful relationships.

Further, future studies need to clarify the mechanisms that lead lonely individuals to be more trusting despite their negative expectations of others. One possibility is that by showing higher trust levels, lonely individuals might want to deliver a clearer signal of trustworthiness to others (Alarcon et al., 2017). They might, for instance, assume that other people have similar negative expectations as themselves and try to compensate by showing higher trust levels in the attempt to reduce other people's uncertainty about their own trustworthiness. Another possible explanation relates to the fact that due to their negative expectations, lonelier individuals might more strongly fear a betrayal by the partner and attempt to coax reciprocation by increasing feelings of guilt (King-Casas et al., 2005). Hence, by sharing more, they might hope to put the partner in an emotionally uncomfortable situation that will increase the likelihood of the partner's cooperation and alleviate their own fear of a betrayal.

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Also, here we studied the relationships of loneliness with trust and trustworthiness expectations as these are the fundaments of humans' whole social fabric and essential components for the establishment of meaningful social relationships—severely disrupted by loneliness. However, the internal motivation lonely individuals have to connect with others might further influence their general generosity and prosociality. Future studies are needed to understand whether and how loneliness impacts prosocial behaviours and how its negative effects on social expectations might hinder their prosociality—thereby impeding a whole series of positive effects that generosity has on a person's well-being (Dunn et al., 2008).

Taken together, these studies demonstrate the opposing effects of loneliness on social behaviours and expectations paving the way for a better understanding of the role of social expectations and behaviours in loneliness, which could help not only explain how feelings of loneliness arise, but also predict the onset of psychopathologies associated with prolonged feelings of loneliness (Porcelli et al., 2019). Further, the unearthed role of personality traits like extraversion in the relationships of loneliness with trusting behaviours and trustworthiness expectations could help develop better, individual-targeted local policies to tackle loneliness, as the common approaches currently in place mostly focus on creating common spaces and social gatherings for people to meet, disregarding individual differences in the appropriateness of these measures.

AUTHOR CONTRIBUTIONS

Gabriele Bellucci: Conceptualization; investigation; methodology; writing – original draft; formal analysis; visualization. **Soyoung Q. Park:** Supervision; conceptualization; funding acquisition; writing – review and editing.

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CONFLICT OF INTEREST STATEMENT

The authors declare no conflicts of interest, financial or otherwise.

DATA AVAILABILITY STATEMENT

The data and code used for the analyses in this paper are made publicly available on a public repository (The Open Science Framework) at the following link: https://osf.io/rn4q6/.

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